Considering the Influence of Exotic Structures on the Cosmic Microwave Background

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Abstract

(this is not my actual abstract) My thesis will be on adding modifications to the comic microwave background's angular power spectrum. There is lots of literature on going from a regular power spectrum $P(k)$ of the early universe's fluctuations to the observed C_ℓ spectrum that we can now observe well. The primordial fluctuations, captured by $P(k)$, are taken to be a power law characterized by an amplitude A_s and a scalar spectral index n_s. I will add effects to this $P(k)$ from variety of objects, such as random circles, random straight lines, and concentric circles. These added effects will come with a few parameters whose optimal values I will determine by minimizing the error between the resulting C_ℓ spectrum model, which I will calculate using the `CAMB` Python package, and observed CMB data from the Planck satellite. Once this is done, I will make comments on the presence or absence (very likely the absence) of these added features in the universe and the scales that one would need to probe today in order to determine if they exist or not.

1 Introduction

(this is very rough)

Following the first measurements of the cosmic microwave background, as is explained in Section 2.2, many multi-million dollar observatories and telescopes have been built for the purpose of offering refined measurements of the universe's first release of radiation. This observational data most importantly helps constrain models of the early universe and explain the nature of our universe.

A fairly simple model for the early universe is most commonly used to match the data, which we are continuing to offer more refined measurements of, and so more complex features in the CMB aren't typically given much observational attention, for science always chooses a simpler mechanism over a more complicated one that both explain the same data. One can add additional parameters to a model at will, but a correspondingly greater fit to observational data that these additional, fine-tuned parameters gives should never be considered a demonstration that a more accurate model for the system of interest has been developed. In general, one needs a great deal of evidence to warrant the adding of parameters into a model that already explains the data at hand.

Our current observations of the radiation across the field of the cosmic microwave background seems, to great accuracy, to be very uniform and structureless; that one point on the field will have almost zero statistical correlation with another point randomly chosen. As will be explained [1, 2]

Briefly summarize the aim of the thesis and how all of the parts in the background material will fit together.

2 Background Material

2.1 Correlation Functions and Power Spectra

Two point correlation function $\xi_2(\Delta \boldsymbol{r})$ between two points $\boldsymbol{r}$ and $\boldsymbol{r}'$ is defined as the expectation value of the field evaluated at two separate points;

$$\xi_2(\Delta \boldsymbol{r}) = \langle \phi(\boldsymbol{r})\phi(\boldsymbol{r} + \Delta \boldsymbol{r}) \rangle \,. \tag{1}$$

Given some point $\boldsymbol{r}_0$ in a field $\phi(\boldsymbol{r})$, the two-point correlation function tells you how likely you are to find the same field value at a separate point $\boldsymbol{r}_0 + \Delta \boldsymbol{r}$. For a completely uncorrelated field with mean field value $\bar{\phi}$, the correlation function becomes[1]

$$\xi_2(|\boldsymbol{r}_i - \boldsymbol{r}_j|) = \langle \phi(\boldsymbol{r}_i)\phi(\boldsymbol{r}_j) \rangle = \delta_{ij} - \bar{\phi}^2.$$

In other words, knowing $\phi(\boldsymbol{r}_i)$ tells us no information about $\phi(\boldsymbol{r}_j)$, which is the definition of an uncorrelated, or a random, field.

In contrast, the two-point correlation function for a completely uniform field ϕ_0 just returns zero, since any point on the field has no statistical correlation to any other point.

The two-point correlation function, expressed in Eq. (1), can be defined using the expectation value's spatial integral, which gives us

$$\xi_2(\Delta \boldsymbol{r}) = \frac{1}{V} \int_V \phi(\boldsymbol{r})\phi(\boldsymbol{r} + \Delta \boldsymbol{r}) \; dx. \tag{2}$$

If we are given a correlation function for a field and instead want to arrive at its power spectrum, which is a tool that tells us how a field is distributed over frequency, we take the

[1] See Appendix A for the derivation.

inverse Fourier transform of (2) to find[2]

$$P(k) = \frac{1}{(2\pi)^3} \int \xi_2(\boldsymbol{r}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}} d^3r. \tag{3}$$

This relationship between a given corelation function and its power spectrum will be crucial to the work of this thesis, as it will allow us to arrive at power spectra from known correlation functions.

2.2 Cosmological Physics and the Cosmic Microwave Background Radiation

The theory of cosmic inflation, the leading class of models that describe the very early phase of our universe's development[3], was originally posed in 1981 [3] as an alternative to the standard big bang model, motivated by the standard theory's flaws that become present when one extrapolates it back to very early times. Inflationary cosmology posits that the early universe underwent a period of exponential expansion driven by the dynamics of a scalar field $\phi(\boldsymbol{r}, t)$, called the inflaton. Under inflation, the scale of the universe increased by an enormous amount during this very early period, expanding by a factor of 10^{50} times greater than what is predicted by standard cosmology [7].

The inflaton field $\phi(\boldsymbol{x}, t)$ began as a quantum field, subject to quantum fluctuations, whose very nature gave rise to this exponential expansion. Quantum fluctuations normally exist on imperceptible scales, but through the mechanism of inflation, these primordial fluctuations were stretched to astronomical scales, eventually being "frozen in" as classical density fluctuations $\delta\rho(\boldsymbol{x})$ which formed the seeds for cosmic structure and the later development of life.

The first light released from the universe, called the cosmic microwave background radiation (CMB), was first serendipitously discovered in 1964 by Arno Penzias and Robert

[2]The derivation of Eq. (3) is given in Appendix B.

[3]See [3, 4, 5, 6] for foundational papers in the theory of inflation.

Wilson [8] at Bell Telephone Laboratories. Many telescopes and observational facilities have since been built to more precisely measure this radiation, with the first, called the Cosmic Microwave Background Explorer (COBE), telescope launched in 1989 [9]. Although Penzias and Wilson's first observation of the CMB revealed a near perfectly uniform field of radiation, our continued precision in observational measurement of this light has slowly revealed its energy's non-uniformity. Fig. 1 demonstrates how the increased precision in CMB measurements have revealed this field's non-uniformity.

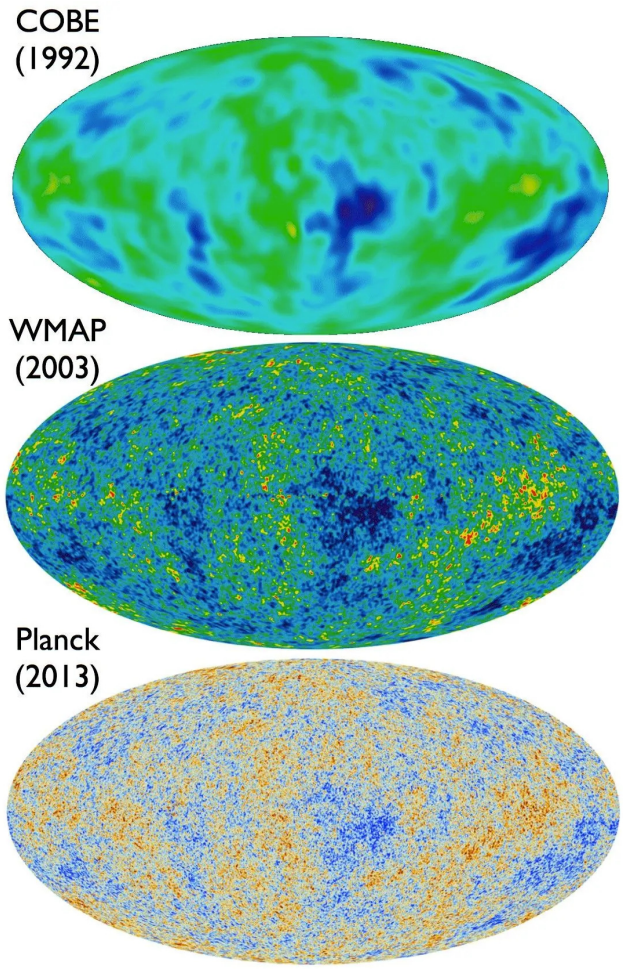

Figure 1: The cosmic microwave background field as measured by COBE in 1990 [10], WMAP in 2003 [11], and the Planck satellite in 2018 [12] demonstrate the slow emergence of this field's non-uniformity of energy. This non-uniformity in energy is captured by measuring the field's temperature, which corresponds to the color given in these plots. Note these maps are spherical, meaning the CMB radiation is the 2D surface formed from a sphere centered around the Earth. These maps are analogous to how the Mercator map project's the Earth's geography onto a 2D plot.

This non-uniformity of the CMB is subject to a great deal of attention in physics and astrophysics, as these fluctuations were ultimately caused by the quantum mechanical nature of the early universe, which has statistical uncertainly inherently built into its nature. As

such, the CMB is our only current observational probe of the early state and development of the universe[4]. When one creates models for the CMB, assumptions of the nature of the early universe's radiation field, and ultimately the nature of inflation, are therefore baked into the models.

To model the CMB, it is standard for one to plot the angular power spectrum C_ℓ as a function of multipole moments ℓ, which describes how the radiation is distributed over different angular scales. The multipole moments ℓ are the angular analog of wavenumbers k, which tell you how many ripples fit on a 360° around the sky.

In this thesis, we will be generating power spectra from known correlation functions. In order to introduce the influence of these power spectra, which live in flat Euclidean space, on the *angular* power spectrum of the CMB, we must use the relationship

$$C_\ell = \frac{4\pi}{(2\pi)^3} \int P(k) j_\ell^2(kr)\ dk, \tag{4}$$

which tells us how to project a power spectrum from flat space onto the surface of a spherical object. This projection is derived in Appendix C.

In standard inflationary cosmology, the power spectrum that generates the CMB fluctuations, called the primordial power spectrum and denoted as $\mathcal{P}(k)$, is modeled as a power law with free amplitude A and spectral index n_s parameters:

$$\mathcal{P}(k) = A_s \left(\frac{k}{k_0}\right)^{1-n_s}. \tag{5}$$

Intuitively, this power law tells us that **(what? That there is no clearly defined structure?)**. From modern CMB observations, we have determined $A_s = 2.09681 \times 10^{-9}$ and $n_s = .965$ [15].

(Briefly explain how this $P(k)$ leads to a uniform CMB field... and state what

[4]Though there has been work on what new information and constraints the observation of gravitational waves from this era, called primordial gravitational waves in the literature, would bring. For example, see [13, 14]

the correlation function is for this case here)

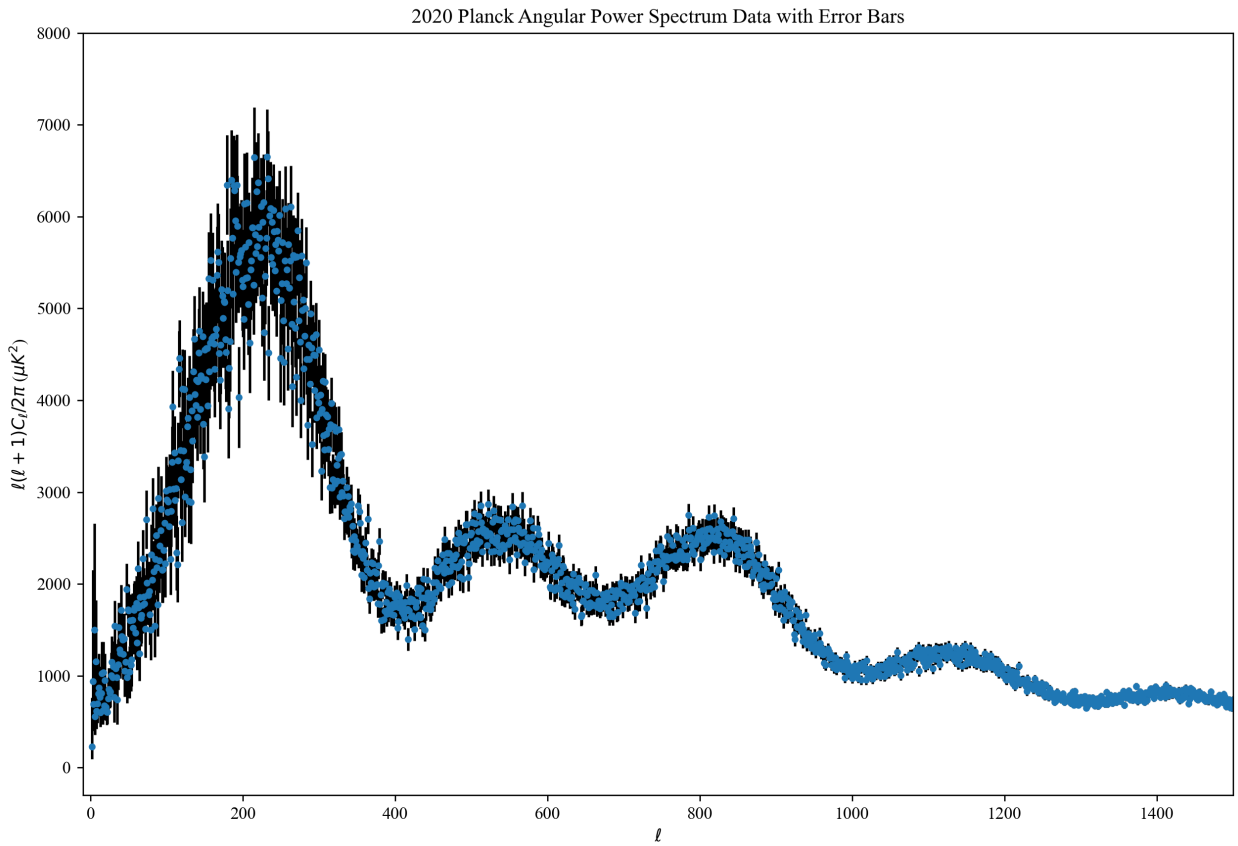

Figure 2:

2.3 Bayesian Analysis

The Bayesian interpretation of statistics, as opposed to frequentist statistics, states that the probability of a variable of interest quantifies our uncertainty about it, rather than representing the frequency of specific values of this variable that is taken on. Since probability is a variable's uncertainty under the Bayesian philosophy, it is said to represent a measure of our belief about this variable. In probability being a measure of our knowledge of a variable, it follows that probability is *not* only relies on the result of repeated experiments in which the variable's value is observed, but also includes the effect of what we believe the variable's value to be. In this way, Bayesian statistics allows one to fluidly update their knowledge of

a variable upon multiple observations of data that constrain the variable's range of possible values.

Because we often have theoretical predictions for variables of interest in physics, it is very common to employ Bayesian methods that make use of these, since they often constrain the space of parameter values. The framework of Bayesian analysis results in a final probability distribution for variables of interest, which is additionally of great interest to physicists, and is another reason why Bayesian statistics is found widely in physics, especially in contexts where observational data is incoming, and models needs to be constantly updated in accordance with this continuing stream of data.

Bayes theorem, the foundation of Bayesian statistics, states that

$$P(\Theta|D) = \frac{P(\Theta)P(D|\Theta)}{P(D)}. \tag{6}$$

Where $P(\Theta)$, called the *prior* distribution, represents our prior hypothesis about our parameters of interest Θ. When we have no knowledge about our parameters Θ, a constant prior distribution is chosen since this equally weights all possible parameter values.

$P(D|\Theta)$ is called the *likelihood*, which represents the probability of generating the observed data D given model parameters Θ and is often written as $\mathcal{L}(\Theta)$[5].When measurements made to construct the data are made up of random, independent samples, the central limit theorem is used to state that the likelihood for these processes should be a Gaussian distribution;

$$\mathcal{L}(\Theta) \propto \exp\left(-\sum_i \frac{\left(D_i - M_i\right)^2}{2\sigma_i^2}\right), \tag{7}$$

$P(D)$ is called the *evidence*, which represents the probability of seeing the data under all possible parameter values;

$$P(D) = \int P(D|\Theta)P(\Theta)d\Theta. \tag{8}$$

[5]It's crucial to state that the likelihood is *not* a probability distribution for parameter values of Θ. It is a function $f(\Theta)$ that quantifies how well the parameters describe the observed data.

Because the Planck data provides asymmetric bounds on the uncertainty in C_ℓ, which is the parameter we ultimately seek to fit, we compute the variance in evidence as

$$\sigma_i^2 = \frac{(+\delta\ell_i)^2 + (-\delta\ell_i)^2}{2}.$$

The evidence acts as a normalizing factor and is typically ignored in techniques that sample from the likelihood and prior, such as Markov chain Monte Carlo (MCMC) techniques. Sampling techniques are especially useful when the model parameters make up a high dimensional space, which makes the integral in Eq. (8) very computationally expensive.

The result of Eq. (6), the *posterior* distribution $P(\Theta|D)$, represents our updated hypothesis about the parameters of interest upon observation of the data. Achieving a distribution for each parameter is the goal of Bayesian statistics, for this tells us which parameter values most likely fit the given data. When the posterior cannot be directly evaluated, it is often approximated using MCMC sampling.

In greater detail, Monte Carlo methods approximate a distribution by using a Markov chain to sample from it. The Markov chain takes a stochastic walk around the distribution of interest, and therefore approximates its behavior without needing a full analytic expression for it. This Markov chain Monte Carlo sampling technique is used to approximate distributions whose behavior is too difficult to represent analytically or computationally expensive to do so. In the context of Bayesian analysis, MCMC methods are used to approximate the posterior as $P(\Theta|D) \approx P(\Theta)P(D|\Theta)$ since the evidence integral, given in Eq. (8), can easily become difficult to evaluate given that integrates over all allowed Θ values.

We are allowed to approximate the posterior using MCMC methods, which drop the evidence term, because the sampling technique relies on *relative probability*; how the probability of a randomly chosen point in the $P(\Theta|D)$ distribution looks *compared* to the subsequently chosen values throughout the sampling process whereas in the full expression of Bayes theorem in Eq. (6), $P(D)$ acts as a normalization constant, so the posterior's probabilities are

normalized and can be compared in an *absolute* sense.

The final statistical tool to be introduced will allow us to quantify the support for our models that consider the influence of additional structure in the CMB correlation function. In considering models that introduce new parameters to fit data that is already well-fit by more simple models, one must address whether or not the additional parameters are providing any additional fit that can't be explained by just the increased degrees of freedom in the model. This problem of overfitting one's data is a common concern in physics, and the necessary skepticism towards high-parameters models is best captured in John Von Neumann's phase, as told by Enrico Fermi [16],

> *"With four parameters I can fit an elephant, and with five I can make him wiggle his trunk."*

To avoid positive conclusions about the ability of the additional structure models to better explain the CMB data, we will use another technique common to Bayesian analysis, called the *Bayes factor*. The Bayes factor takes the ratio of two competing models' likelihoods, which will be the standard Lambda cold dark matter model of cosmology and the added structure models in our case. The Bayes factor therefore looks like

$$K = \frac{P(D|\Theta_{\text{mod}})}{P(D|\Theta_{\Lambda CDM})}. \tag{9}$$

If the added structure models *do* result in a better fit that is actually meaningful, the Bayes factor value will be a large number. Any $K > 1$ indicates that Θ_{mod} will be more preferred by the data than $\Theta_{\Lambda CDM}$. The strength of this support for a given K value is given here

K **Value**	**Strength of Evidence**
$1 - 3.2$	Insignificant
$3.2 - 10$	Substantial
$10 - 100$	Strong
> 100	Decisive

Table 1: Bayes factor confidence levels sorted by their support for Θ_{mod}'s explanatory power as compared to $\Theta_{\Lambda CDM}$.

Question to Dr. Dahlstrom: do you think I should change the order of topics in Bayesian statistics? I currently have it as (1) provide motivation and define Bayes's theorem, (2) explain MCMC sampling, (3) explain Bayes factor... I'm wondering if I should move MCMC sampling somewhere else.

3 Procedures

This thesis investigates the influence of additional structure on the primordial power spectrum by comparing the resulting C_ℓ spectrum that's generated with observational data from the Planck satellite, which is openly provided by the Planck Legacy Archive. The two-point correlation function has been derived for a variety of random fields, such as those of lines and concentric circles in [17]. Given these correlation functions, I first derive their corresponding $P(k)$. I then construct a total modified primordial power spectrum $P'(k) = P(k) + \mathcal{P}(k)$ where $\mathcal{P}(k)$ is the standard primordial power spectrum given in Eq. (5), and use the Code for Anisotropies in the Microwave Background (`CAMB`) Python package [18] to generate resulting modified angular power spectra C'_ℓ. Given the set of modified angular power spectra that result from this procedure for each added $\xi_2(\Delta r)$, I then plot these against the spectrum generated by standard power-law inflationary cosmology in regards to their fit with Planck's data.

To determine the goodness-of-fit for these modifications, I first carry out a Bayesian analysis on the free parameters of each specific model to find their optimal values. With these, I then compute a Bayes factor with the power law data to quantify how likely the modified models explain the observed CMB data.

To perform Bayesian analysis with the goal of finding the optimal values for the modified spectrum parameters, which are the density of circles ρ and their radii R in the case of the identical circle field, we assume uniform priors on both of these parameters over a range **(I need to justify why the range I'm using is good)** since we have no prior knowledge of what value these should take. We also assume the likelihood follows the exponential distribution given in Eq. (7) because CMB measurements are taken randomly and independently.

Because the likelihood and priors will be computationally evaluated in Python, we use a MCMC sampler rather than additionally computing the evidence integral, as explained in the background section on Bayesian statistics. In this work, I use the `emcee` Python package to carry out the sampling portion of this work.

To determine how likely these modifications to the primordial power spectrum better explain the angular power spectrum of the CMB, I finally compute the Bayes factor, as defined in Eq. (9).

3.1 Uniform Radius Random Circle Field

Take the two-point correlation function for a random field of identical circles as derived in [17], given as

$$\xi_2(r) = \frac{1}{\pi \rho r \sqrt{1 - \left(\frac{r}{2R}\right)^2}} \tag{10}$$

where ρ is the density of these circles and R is their fixed radius. Using Eq. (3) to compute the power spectrum of this field, we find that

$$P_\circ(k) = \frac{1}{(2\pi)^3}\int_0^{2R} \frac{e^{ikr}}{\pi\rho r\sqrt{1-\left(\frac{r}{2R}\right)^2}}dr = \frac{R}{2\pi k\rho}J_0(kR)^2 \tag{11}$$

where the integral's bounds come from the range of validity for the correlation function to hold.

The uniform radius random circle field's power spectrum plotted on scales relevant for the CMB is compared to the standard power law spectrum, Eq. (5), in Fig. 3.

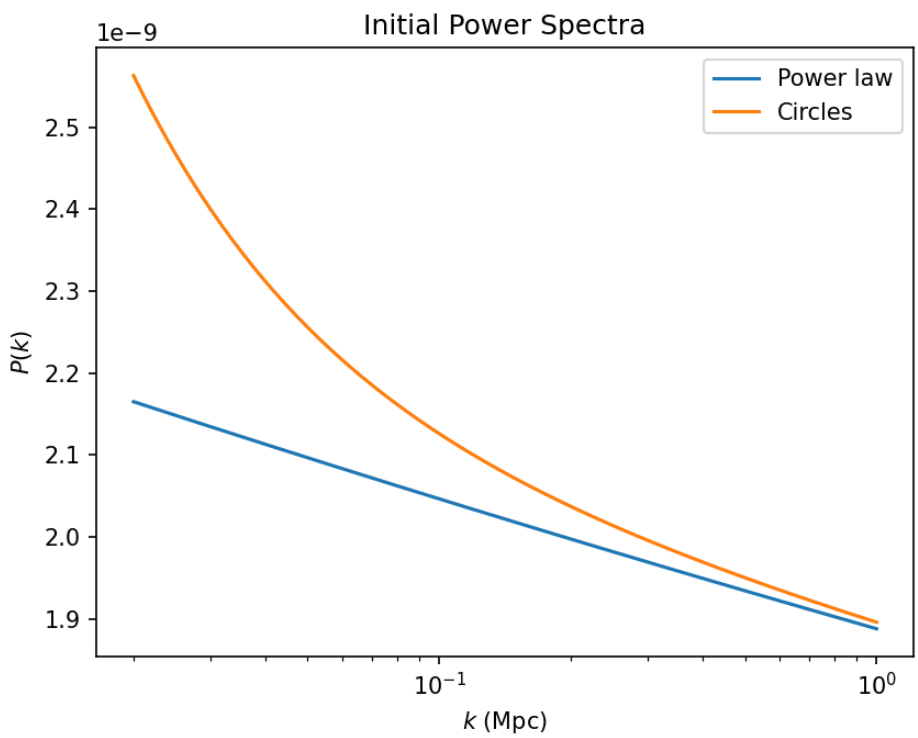

Figure 3: The power spectrum generated by the uniform radius random circle field modification compared against the typical cosmological power law spectrum. For future importance, note the significant deviation at small k values.

To implement this modification into the `CAMB` code, we write the full power spectrum as a

sum of the standard power law and the result of Eq. (11),

$$P'(k) = \mathcal{P}(k) + \frac{R}{2\pi k\rho} J_0(kR)^2, \tag{12}$$

and input this into the `set_initial_power_function` function. From here, we carry out our analysis as detailed in the following section.

4 Results

With the additional correlation function structure, we now wish to determine how this compares against the observed CMB data. Given a primordial power spectrum, values for the Hubble constant H_0, the baryon density parameter $\Omega_b h^2$, the cold dark matter density parameter $\Omega_c h^2$, and the reionization optical depth τ, `CAMB` will generate an angular power spectrum C_ℓ plot that can be directly compared against the observed CMB data from the Planck satellite.

Question 2 for Dr. Dahlstrom: I haven't defined any of the parameters mentioned above. Should I also define these in the section 2.2? Or the Procedures section where I am directly talking about the CAMB code?

We therefore generate a C_ℓ from modified power spectrum from a random field of uniform radius circles, which is plotted in Fig. 4 for randomly-chosen R and ρ values which parameterize the field.

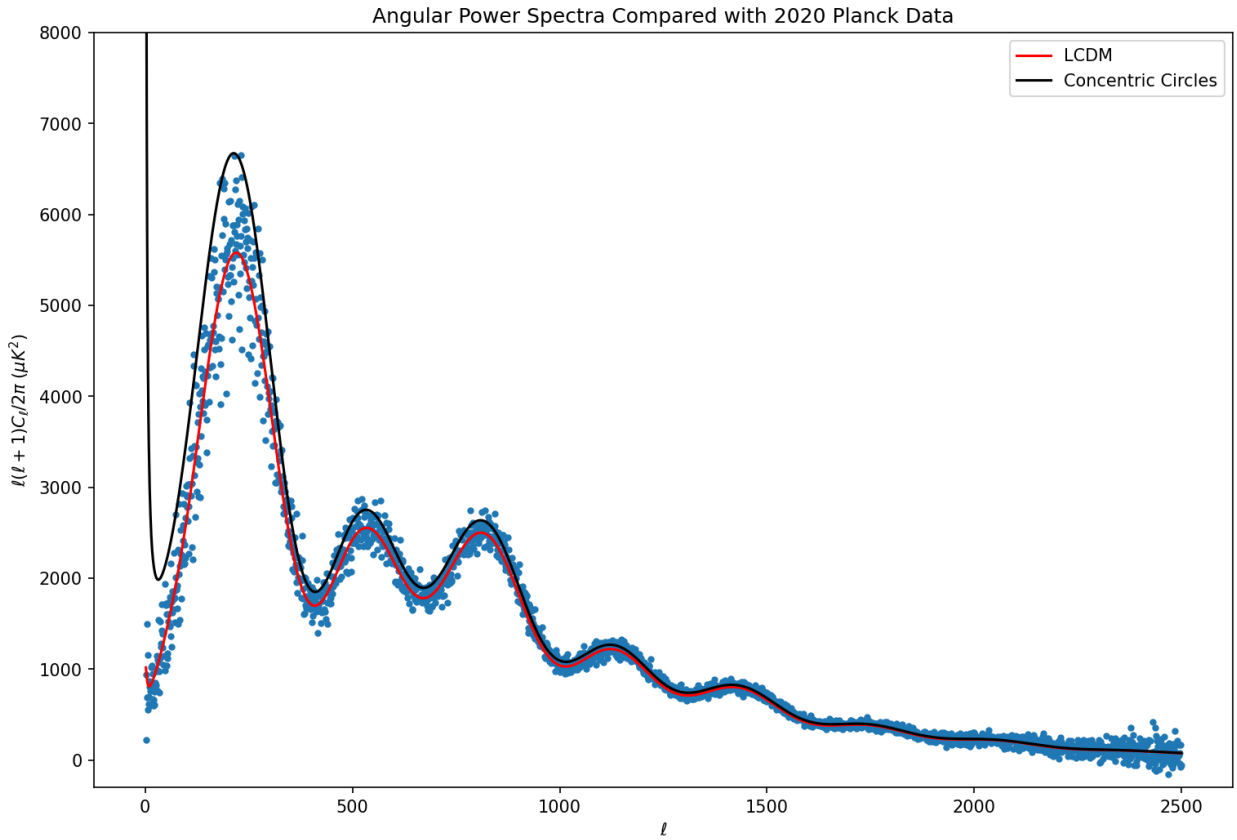

Figure 4: Model with nonideal, randomly chosen parameters $R = 5 \times 10^{-9} m$ and $\rho = 100$. **(have to figure out the units on this)**

By using the Bayesian analysis method outlines in Section 2.3, we construct a posterior distributions for R and ρ and generate a corner plot to examine these distributions in Fig. 5.

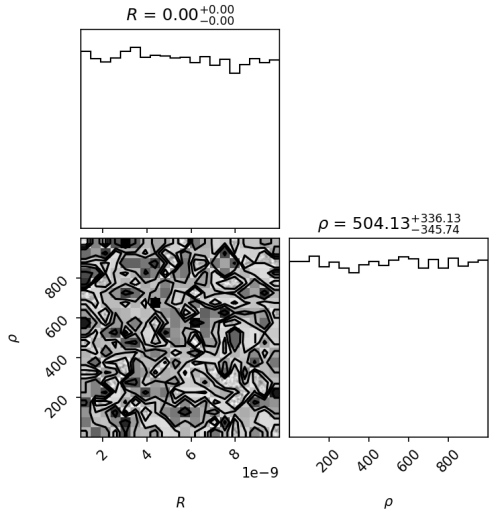

Figure 5: Posterior distributions for R and ρ in the uniform-radius random field modification. **(I know this is really bad. I'm working on fixing my method. I will also have this display the optimal value)**

From the posterior distributions for R and ρ, we take the most probable value for each to again plot our uniform-radius random field modification model against the LCDM model and the observed CMB data to proceed with our analysis. Doing so results in Fig. 6

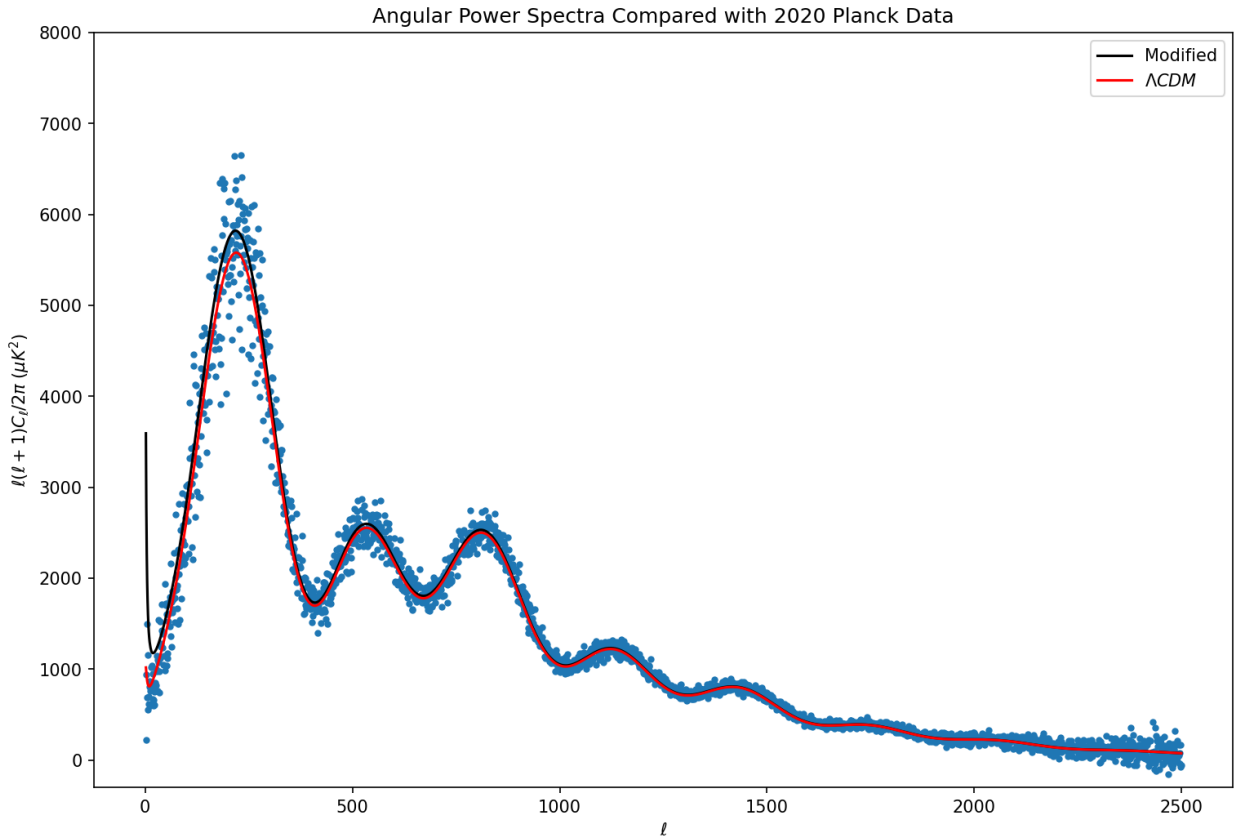

Figure 6: Model with optimized $R = 5.41 \times 10^{-9} \pm 2.6 \times 10^{-9}$ and $\rho = 504.135 \pm 288.99$ values plotted against LCDM model and Planck CMB C_ℓ data. **(I again know these errors are awful... I'm working on it)**

We now plot the difference between our optimized uniform-radius random field model and the standard LCDM model over multipole moments ℓ in Fig. 7.

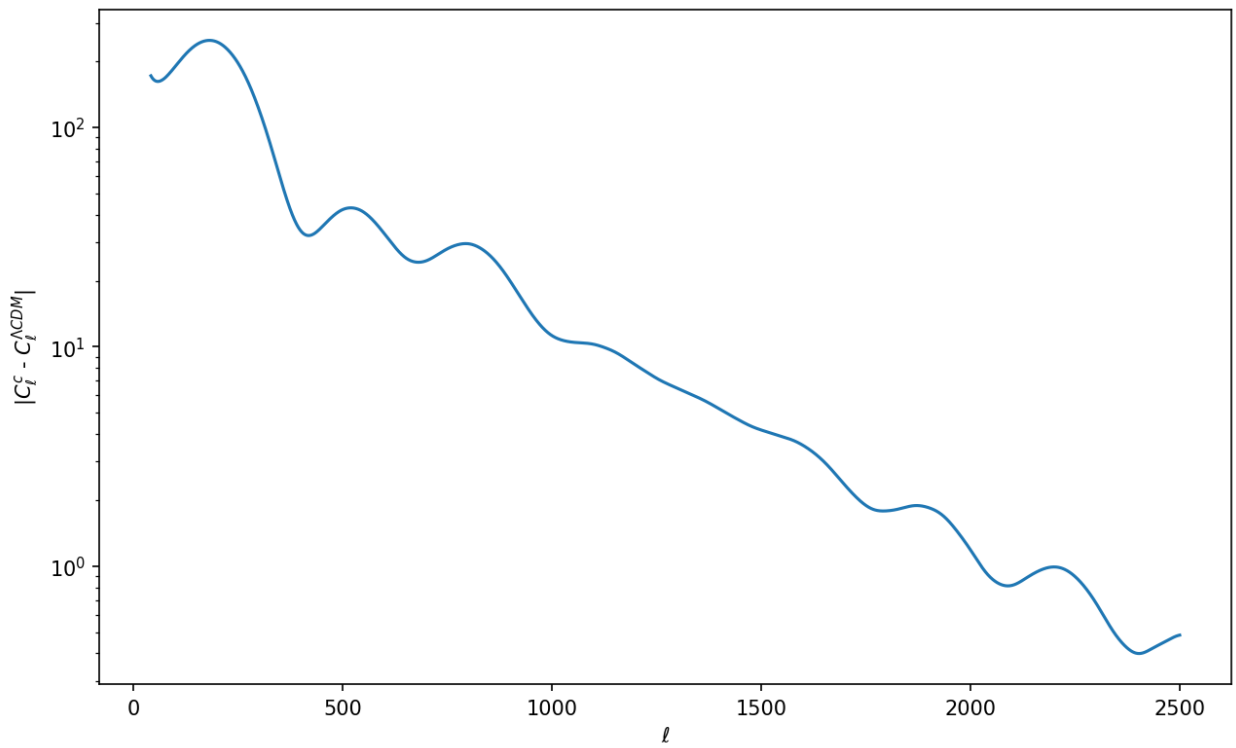

Figure 7: The difference between C_ℓ generated by the optimized uniform-radius random field model and the standard LCDM model. The significant deviations at low ℓ can be easily seen in Fig 6, but the smaller deviations at high ℓ are not present by eye.

To quantify the alignment of both models to the Planck data, we compute a χ^2 statistic for both models, which is the exponential term of Eq. (7). In doing so, we find that

$$\frac{\chi^2_{\text{mod}}}{\chi^2_{\Lambda CDM}} = 1.15, \tag{13}$$

which tells us that the standard lambda cold dark matter model fits the data better than our optimized uniform radius random circle field modification **(this is because of the blow up at low ℓ values in my model. I will fix this and explain how I do this in the Procedures. I will proceed assuming that chisq will flip when this is fixed)**.

5 Discussion

Given that our χ^2 statistic in Eq. (13) tells us that our modified model fits the data better than the standard cosmological model, we now wish to determine whether or not this additional agreement is just the result of the addition of more free parameters. To do so, we apply the Bayes factor, as defined in Eq. (9) of Section 2.3. In computing the Bayes factor between our two competing models, we find $K = 2.74$, which tells us that the better fit provided by our modified model is insignificant, according to Table 1.

The low Bayes factor value combined with the low gain in χ^2 that our modification allows allows us to conclude that, although the uniform radius random circle field provides a small amount of additional fit to the observed CMB data compared to the standard cosmological model, this gain in fit does not justify the incorporation of two additional parameters. This tells us that the addition of the R and ρ parameters in our model do nothing more than over fit the data, which we set out to avoid.

6 Conclusion

Question 3 to Dr. Dahlstrom: you want future directions discussion in the Conclusions section, right?

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Based on observations obtained with Planck (**??**), an ESA science mission with instruments and contributions directly funded by ESA Member States, NASA, and Canada.

A Two-Point Correlation Function of Uncorrelated Field

Using Eq. (2)

$$\xi_2(|\boldsymbol{r}_i - \boldsymbol{r}_j|) = \frac{1}{V}\int(\phi(\boldsymbol{r}_i) - \bar{\phi})(\phi(\boldsymbol{r}_j) - \bar{\phi})d^3x$$

When this is expanded and the fact that integrating over the whole field, scaled by its volume returns the mean field value;

$$\bar{\phi} := \frac{1}{V}\int \phi(\boldsymbol{r}_i)d^3x,$$

we get

$$\xi_2(|\boldsymbol{r}_i - \boldsymbol{r}_j|) = \frac{1}{V}\int \phi(\boldsymbol{r}_i)\phi(\boldsymbol{r}_j)d^3x - \bar{\phi}^2. \tag{14}$$

Because the field is uncorrelated, the expectation value of the field at two separate points will always return zero, giving us

$$\xi_2(|\boldsymbol{r}_i - \boldsymbol{r}_j|) = \delta_{ij} - \bar{\phi}^2.$$

B Fourier Transform of Two-Point Correlation Function:

Take Fourier transform

$$\phi(\boldsymbol{r}) = \frac{1}{(2\pi)^3}\int \phi(\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}d^3k \tag{15}$$

So if we define $\boldsymbol{r}' = \boldsymbol{r} + \Delta\boldsymbol{r}$, we get

$$\xi_2(\Delta\boldsymbol{r}) = \langle\phi(\boldsymbol{r})\phi(\boldsymbol{r}+\Delta\boldsymbol{r})\rangle = \left\langle \frac{1}{(2\pi)^3}\int \phi(\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}d^3k\frac{1}{(2\pi)^3}\int \phi(\boldsymbol{k}')e^{-i\boldsymbol{k}'\cdot(\boldsymbol{r}+\Delta\boldsymbol{r})}d^3k' \right\rangle \tag{16}$$

using the fact that $\xi_2(\Delta\boldsymbol{r})$ is a real-valued function, which requires the complex conjugate

to be taken of the $\mathscr{F}\{\phi(\boldsymbol{r}+\Delta\boldsymbol{r})\}$ term.

Using the Wiener-Khinchin Theorem for random variables in 3D, $\langle\phi(\boldsymbol{k})\phi(\boldsymbol{k}')\rangle = (2\pi)^3\delta^3(\boldsymbol{k}-\boldsymbol{k}')P(k)$ [19], we get

$$\xi_2(\Delta\boldsymbol{r}) = \frac{1}{(2\pi)^3}\int\int \delta^3(\boldsymbol{k}-\boldsymbol{k}')P(k)e^{i\boldsymbol{k}\cdot\boldsymbol{r}}e^{-i\boldsymbol{k}'\cdot(\boldsymbol{r}+\Delta\boldsymbol{r})}d^3k'd^3k. \tag{17}$$

Integrating over the $\delta^3(\boldsymbol{k}-\boldsymbol{k}')$ term results in

$$\xi_2(\Delta\boldsymbol{r}) = \frac{1}{(2\pi)^3}\int P(k)e^{-i\boldsymbol{k}\cdot\Delta\boldsymbol{r}}d^3k. \tag{18}$$

Finally taking the inverse Fourier transform of this expression, we're left with

$$P(k) = \frac{1}{(2\pi)^3}\int \xi_2(\boldsymbol{r})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}d^3r, \tag{19}$$

which tells us how to arrive at a powr spectrum for a field given its correlation function.

C Deriving the Angular Power Spectrum from Initial Fluctuations

$$a_{\ell m} = \int \phi(\boldsymbol{r})Y^*_{\ell m}(\hat{n})d\Omega \tag{20}$$

Using the Fourier transform of $\phi(\boldsymbol{r})$

$$a_{\ell m} = \frac{1}{(2\pi)^3}\int\int \phi(\boldsymbol{k})Y^*_{\ell m}(\hat{n})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\ d^3k\ d\Omega \tag{21}$$

Using the plane wave expansion:

$$e^{i\boldsymbol{k}\cdot\boldsymbol{r}} = 4\pi\sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} i^\ell j_\ell(\boldsymbol{k}\cdot\boldsymbol{r})Y_{\ell m}(\hat{\boldsymbol{r}})Y^*_{\ell m}(\hat{\boldsymbol{k}}) \tag{22}$$

where the spherical Bessel function $j_\ell(kr)$ is defined as:

$$j_\ell(kr) = \sqrt{\frac{\pi}{2kr}} J_{\ell+1/2}(kr) \tag{23}$$

$$a_{\ell m} = \frac{4\pi}{(2\pi)^3} \int\int \phi(\boldsymbol{k}) Y^*_{\ell m}(\hat{n}) \sum_{\ell'=0}^{\infty} \sum_{m=-\ell'}^{\ell'} i^{\ell'} j_{\ell'}(\boldsymbol{k}\cdot\boldsymbol{r}) Y_{\ell' m'}(\hat{\boldsymbol{r}}) Y^*_{\ell' m'}(\hat{\boldsymbol{k}})\ d^3k\ d\Omega \tag{24}$$

Using the fact that

$$\int Y^*_{\ell m}(\hat{n}) Y_{\ell' m'}(\hat{n}) d\Omega = \delta_{\ell\ell'}\delta_{mm'}, \tag{25}$$

this gives us

$$a_{\ell m} = \frac{4\pi}{(2\pi)^3} \int \phi(\boldsymbol{k}) \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} i^{\ell} j_{\ell}(\boldsymbol{k}\cdot\boldsymbol{r}) Y^*_{\ell m}(\hat{\boldsymbol{k}})\ d^3k \tag{26}$$

Now take $\langle |a_{\ell m}|^2 \rangle$, using $\langle \phi(\boldsymbol{k})\phi(\boldsymbol{k}') \rangle = (2\pi)^3 \delta^3(\boldsymbol{k}-\boldsymbol{k}') P(k)$:

$$\left\langle |a_{\ell m}|^2 \right\rangle = \frac{4\pi}{(2\pi)^3} \int\int P(k)\delta^3(\boldsymbol{k}-\boldsymbol{k}') \sum_{\ell\ell'} \sum_{mm'} i^{\ell} j_{\ell}(\boldsymbol{k}\cdot\boldsymbol{r}) Y^*_{\ell m}(\hat{\boldsymbol{k}}) i^{\ell'} j_{\ell'}(\boldsymbol{k}'\cdot\boldsymbol{r}) Y_{\ell' m'}(\hat{\boldsymbol{k}'})\ d^3k' d^3k \tag{27}$$

$$\left\langle |a_{\ell m}|^2 \right\rangle = \frac{4\pi}{(2\pi)^3} \int P(k) \sum_{\ell\ell'} \sum_{mm'} i^{\ell} j_{\ell}(\boldsymbol{k}\cdot\boldsymbol{r}) Y^*_{\ell m}(\hat{\boldsymbol{k}}) i^{\ell'} j_{\ell'}(\boldsymbol{k}\cdot\boldsymbol{r}) Y_{\ell' m'}(\hat{\boldsymbol{k}})\ d\Omega dk \tag{28}$$

$$\left\langle |a_{\ell m}|^2 \right\rangle = \frac{4\pi}{(2\pi)^3} \int P(k) \sum_{\ell\ell'} \sum_{mm'} i^{\ell} j_{\ell}(\boldsymbol{k}\cdot\boldsymbol{r}) i^{\ell'} j_{\ell'}(\boldsymbol{k}\cdot\boldsymbol{r}) \delta_{\ell\ell'}\delta_{mm'}\ dk \tag{29}$$

$$\left\langle |a_{\ell m}|^2 \right\rangle = \frac{4\pi}{(2\pi)^3} \int P(k) \sum_{\ell} \sum_{m} i^{2\ell} j^2_{\ell}(\boldsymbol{k}\cdot\boldsymbol{r})\ dk \tag{30}$$

Since $i^{2\ell} = 1$,

$$C_\ell = \frac{4\pi}{(2\pi)^3} \int P(k) j^2_\ell(kr)\ dk \tag{31}$$

This tells us that given a power spectrum $P(k)$, we can generate an angular power spectrum $C(\ell)$ using this integral!